

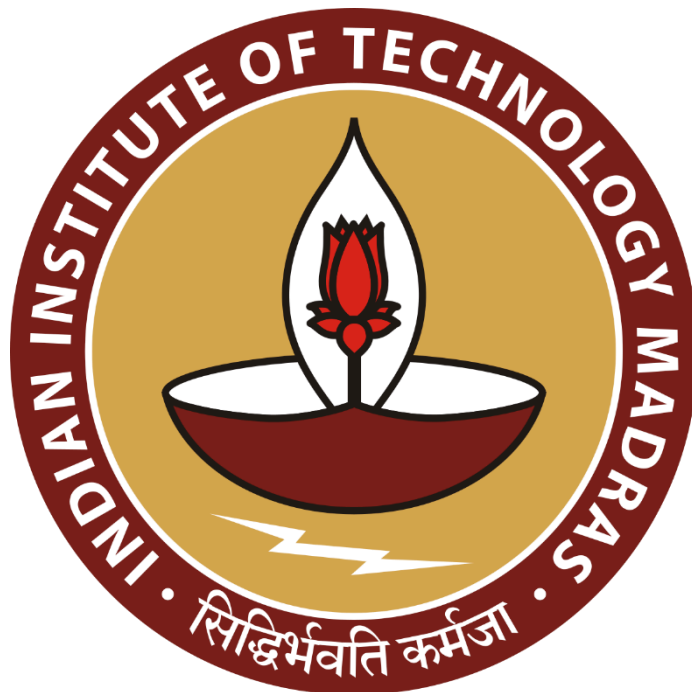
Profit Optimization Using Sales Analytics

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled "Profit Optimization Using Sales Analytics". I extend my appreciation to **TrendyThreads Pvt. Ltd.** , for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:

A handwritten signature in blue ink that reads "Amandeep".

Name: Amandeep Singh

Date: June 12 2025

1 Executive Summary and Title (209 Words)

TrendyThreads Pvt. Ltd. is a B2C retailer in fashion that primarily sells through Amazon India. TrendyThreads sells merchandise across a variety of women's clothing, including ethnic kurtas and dress sets, western tops and blouses. TrendyThreads utilizes merchant fulfilled and fulfilled by Amazon to distribute merchandise across India. The company is consistently getting orders but is struggling with direct profitability due to different operational inefficiencies. Some SKUs have either no sales or \$0 sales, which creates unnecessary inventory holding costs and blocked capital in low margins products. Furthermore, high cancellation rates and return rates undermine associated revenues and present underlying operational and customer satisfaction issues in processes. TrendyThreads must also have insufficient diversified product ranges and is overly reliant on a few product categories which is restricting revenue growth and introduces business risk to any disruptions of those. The objective of the project is to use data analytic techniques to identify trends and patterns from exploratory data analysis, benchmarking SKU performance, fulfilment analysis, and analyzing returns. Based on analysis, TrendyThreads will use findings to identify low-performing SKUs, enhanced operational fulfilment plans, and increased offered product lines of last exemplifying SKU performance. Improvements to top & bottom line are the measures of success to help TrendyThreads adapt and sustainability grow as business.

2. Organization Background (118 Words)

TrendyThreads Pvt. Ltd. is a digital fashion retailer in India offering ethnic and western wear for women at affordable price points through the Amazon India platform. By using a hybrid of Fulfilled by Amazon (FBA) and merchant fulfilled options, TrendyThreads can reach a large B2C audience of customers living in urban and semi-urban locations. TrendyThreads large range of products and aggressive price points have enabled rapid growth in a competitive omni-channel market. However, the company now finds itself in a position of operational challenges with stopping inventory, high return rates and profitability problems as a result of category concentration and fulfillment inefficiencies. TrendyThreads is considering data based solutions to improve inventory management, optimize operation, and become profitably sustainable.

2 Problem Statement (Listed as objectives) (69 Words)

- 2.1 Due to many SKUs experiencing zero or immaterial sales, that inventory sits unsold/unclaimed and leads to capital blockage and storage costs.

- 2.2 The company is cancelling or returning over 14% of its orders causing revenue loss, costs associated with additional logistics, and harming their customer experience.
- 2.3 Sales are heavily weighted towards two categories of product, therefore risking and limiting opportunities for diversifying their product offering and expanding their customer base.

3 Background of the Problem (138 Words)

TrendyThreads Pvt. Ltd. is an online fashion retailer on Amazon, but are not seeing growth in profits despite consistent sales volumes. Our data indicate that they are too reliant on categories (Kurta and Set) with high return and cancellation rates. The returns and cancelations often stem from (a) poor delivery time, and (b) not meeting customer expectations whatsoever. As well, while investigating the inventory planning, we found that they had too many SKUs that they could not sell and ended up costing them extra money by occupying space in their warehouse. The business has not analyzed the returns/customer experience for Amazon vs Merchant fulfillment. In order to address TrendyThreads' issue, and we will develop a 5-week project to analyzes sales and operational data for determining inventory management practices, implement fulfillment strategies, and build sound profitability decision-making through data.

4 Problem Solving Approach (364 Words)

5a. Explanation of the methods

The project will take a quantitative approach - using Exploratory Data Analysis (EDA) to summarize correlation between sales, returns and revenue, and clustering analysis to track key performance indicators on all SKUs - volume, return rates, profit are the most relevant figures. Fulfillment Comparison Analysis will also compare Amazon and Merchant fulfillment against KPI metrics of return percent and successful delivery - representing a proxy for delivery cost. Qualitative analysis will take the form of a logic tree to visualize how to categorize insight, categorically dimension - customer satisfaction, inventory cost and category risk. This categorical framing will provide insights necessary for evaluating operational effectiveness to support strategic choices that improve inventory fulfillment at TrendyThreads Pvt. Ltd for category planning purposes.

5b. Explanation of the intended data collection with Justification:

This project will use an existing Amazon Sales Report dataset that contains order status and number, SKU, fulfillment channel, category, quantity, amount and any promotional tags.

Order Data: This includes order status, channel of fulfillment and service level/shipping speed. This allows us to examine operational delays, returns, and delivery quality.

SKU/Customer Product Data: this is for the product-level performance (i.e. revenue generated and quantity over a number sold), which is important for understanding dead stock and high performers.

Promotion and Pricing Data: this is used to understand if discounts and marketing offers impact margin.

Justification:-

Utilizing this dataset allows for a full read of operations without engaging in additional primary data collection, which can be labor intensive. The record includes all of the critical variables needed to gauge operations at both a strategic (category-level) and tactical (SKU-level) level.

5c. Analysis Tools and Rationale:

Analysis Tools:-

Microsoft Excel / Google Sheets: Used for preliminary data exploration and used for pivot analysis and filtering out bad data.

Python (Pandas, Matplotlib, Seaborn): perform detail data manipulation, data visualizations, trend analysis, and clustering.

Power BI / Tableau: Used to build interactive dashboards to help communicate insights effectively.

Rationale:-

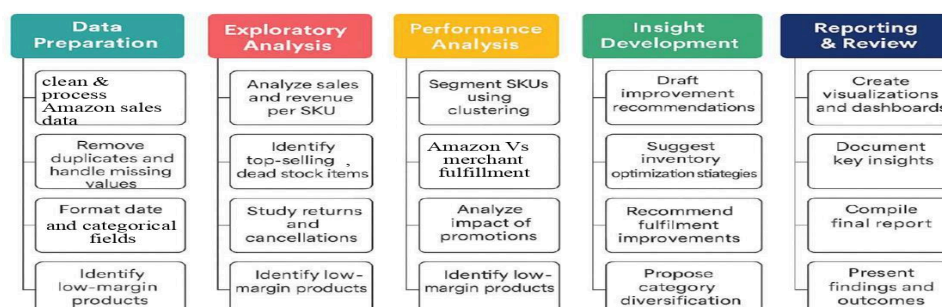
Excel/Sheets: data cleansing, and summarizing data for presentation.

Python: Allows for complex operations, trend discovery, and real-time insights using the libraries pandas, seaborn, scikit-learn.

BI Tools: Great for building visual stories out of data for easily-consumed dashboards, and to get stakeholders.

5 Expected Timeline

5.1 Work Breakdown Structure:



5.2 Gantt Chart:

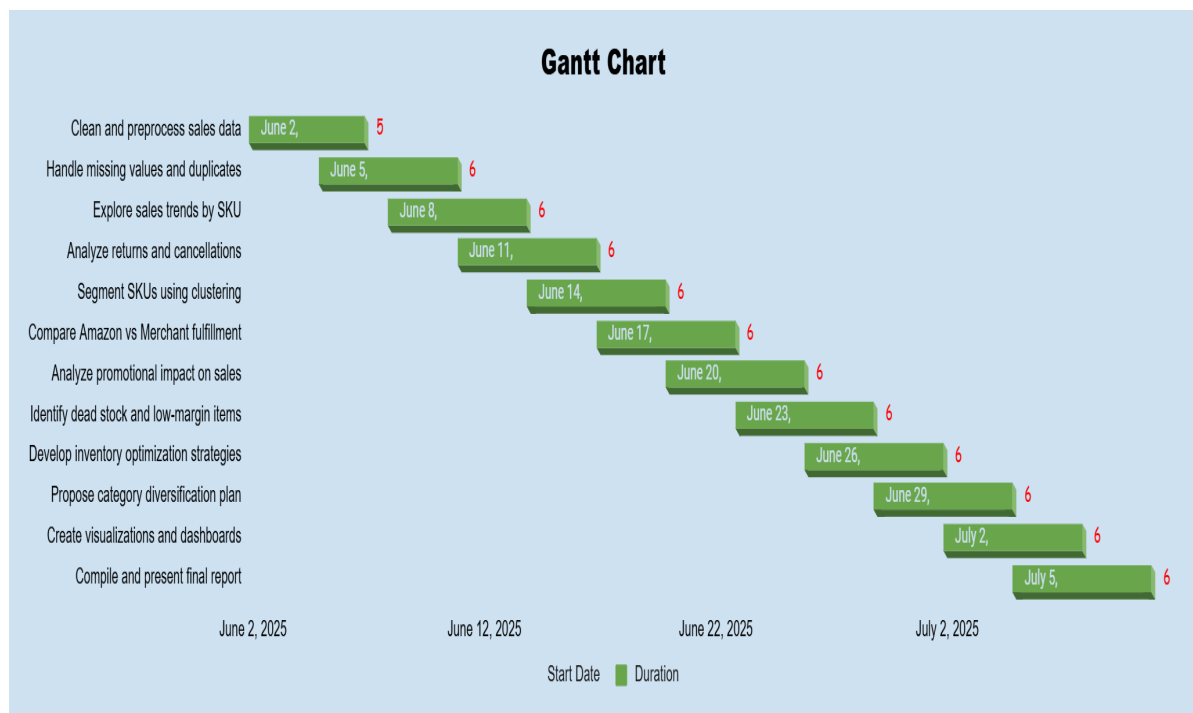


Figure 1 Expected timeline for completion of project.

6 Expected Outcome (158 words)

The project's primary objective is to assist TrendyThreads Pvt. Ltd. in enhancing the quality of operational inventory planning and developing greater visibility into SKUs that are registering a zero or low sales figure. The company can reduce warehouse and logistic fees with better inventory planning. Next, we will examine cancellation and returns patterns. By doing so, the company can identify important issues, as well as product page, size guide, or customer experience improvements to lessen customer dissatisfaction. In addition, the company will analyze Amazon fulfilled and Merchant fulfilled orders to gradually move particular SKUs from MFN to FBA, while reducing delivery errors simultaneously. By clustering product/brand categories based on sales, returns, and margin the project will allow TrendyThreads to better utilize their marketing strategy and discount sales for products in need for improvements. Overall, the project will examine data driven decision making, support a sustain growth improving profitability opportunities, and look for improvements in the e-commerce fashion category.